Lee, Sun, Sun and Taylor (2016, Annals of Statistics)

Master's Thesis Workshop 1

Yasuyuki Matsumura (Kyoto University)

Last Updated: April 16, 2025

https://yasu0704xx.github.io

The Annals of Statistics
2016, Vol. 44, No. 3, 907–927
DOI: 10.1214/15-AOS1371
© Institute of Mathematical Statistics, 2016

EXACT POST-SELECTION INFERENCE, WITH APPLICATION TO THE LASSO

BY JASON D. LEE[*,1], DENNIS L. SUN[+,2], YUEKAI SUN[*,3] AND JONATHAN E. TAYLOR[‡,4]

University of California, Berkeley, California Polytechnic State University*[+] *and Stanford University*[‡]

We develop a general approach to valid inference after model selection. At the core of our framework is a result that characterizes the distribution of a post-selection estimator conditioned on the *selection event*. We specialize the approach to model selection by the lasso to form valid confidence intervals for the selected coefficients and test whether all relevant variables have been included in the model.

Introduction

Post-selection inference in linear regression

Outline of our approach

The lasso and its selection event

Polyhedral conditioning sets

Post-selection intervals for regression coefficients

Data example

Extensions

References

Introduction

Post-selection inference in linear regression

Outline of our approach

The lasso and its selection event

Polyhedral conditioning sets

Post-selection intervals for regression coefficients

Data example

Extensions

References

References i

REFERENCES

BENJAMINI, Y., HELLER, R. and YEKUTIELI, D. (2009). Selective inference in complex research. *Philos. Trans. R. Soc. Lond. Ser. A Math. Phys. Eng. Sci.* **367** 4255–4271. MR2546387

BENJAMINI, Y. and YEKUTIELI, D. (2005). False discovery rate-adjusted multiple confidence intervals for selected parameters. *J. Amer. Statist. Assoc.* **100** 71–93. MR2156820

BERK, R., BROWN, L., BUJA, A., ZHANG, K. and ZHAO, L. (2013). Valid post-selection inference. *Ann. Statist.* **41** 802–837. MR3099122

COX, D. R. (1975). A note on data-splitting for the evaluation of significance levels. *Biometrika* **62** 441–444. MR0378189

EFRON, B., HASTIE, T., JOHNSTONE, I. and TIBSHIRANI, R. (2004). Least angle regression. *Ann. Statist.* **32** 407–499. MR2060166

FISHER, R. (1956). On a test of significance in Pearson's Biometrika Tables (No. 11). *J. Roy. Statist. Soc. Ser. B.* **18** 56–60. MR0082773

FITHIAN, W., SUN, D. and TAYLOR, J. (2014). Optimal inference after model selection. Preprint. Available at arXiv:1410.2597.

JAVANMARD, A. and MONTANARI, A. (2013). Confidence intervals and hypothesis testing for high-dimensional regression. Preprint. Available at arXiv:1306.3171.

KNIGHT, K. and FU, W. (2000). Asymptotics for lasso-type estimators. *Ann. Statist.* **28** 1356–1378. MR1805787

LEEB, H. and PÖTSCHER, B. M. (2005). Model selection and inference: Facts and fiction. *Econometric Theory* **21** 21–59. MR2153856

References ii

LEEB, H. and PÖTSCHER, B. M. (2006). Can one estimate the conditional distribution of post-model-selection estimators? *Ann. Statist.* **34** 2554–2591. MR2291510

LEHMANN, E. L. and ROMANO, J. P. (2005). *Testing Statistical Hypotheses*, 3rd ed. Springer, New York. MR2135927

LOCKHART, R., TAYLOR, J., TIBSHIRANI, R. and TIBSHIRANI, R. (2014). A significance test for the lasso (with discussion). *Ann. Statist.* **42** 413–468. MR3210970

MILLER, A. (2002). *Subset Selection in Regression*, 2nd ed. Chapman & Hall/CRC, Boca Raton, FL. MR2001193

NEGAHBAN, S. N., RAVIKUMAR, P., WAINWRIGHT, M. J. and YU, B. (2012). A unified framework for high-dimensional analysis of M-estimators with decomposable regularizers. *Statist. Sci.* **27** 538–557. MR3025133

PÖTSCHER, B. M. (1991). Effects of model selection on inference. *Econometric Theory* **7** 163–185. MR1128410

PÖTSCHER, B. M. and SCHNEIDER, U. (2010). Confidence sets based on penalized maximum likelihood estimators in Gaussian regression. *Electron. J. Stat.* **4** 334–360. MR2645488

ROBINSON, G. K. (1979). Conditional properties of statistical procedures. *Ann. Statist.* **7** 742–755. MR0532239

SAMPSON, A. R. and SILL, M. W. (2005). Drop-the-losers design: Normal case. *Biom. J.* **47** 257–268. MR2145117

SILL, M. W. and SAMPSON, A. R. (2009). Drop-the-losers design: Binomial case. *Comput. Statist. Data Anal.* **53** 586–595. MR2654594

STOREY, J. D. (2003). The positive false discovery rate: A Bayesian interpretation and the q-value. *Ann. Statist.* **31** 2013–2035. MR2036398

References iii

TAYLOR, J., LOCKHART, R., TIBSHIRANI, R. J. and TIBSHIRANI, R. (2014). Post-selection adaptive inference for least angle regression and the lasso. Preprint. Available at arXiv:1401.3889.

TIAN, X. and TAYLOR, J. (2015). Asymptotics of selective inference. Preprint. Available at arXiv:1501.03588.

TIBSHIRANI, R. (1996). Regression shrinkage and selection via the lasso. *J. Roy. Statist. Soc. Ser. B* **58** 267–288. MR1379242

TIBSHIRANI, R. J. (2013). The lasso problem and uniqueness. *Electron. J. Stat.* **7** 1456–1490. MR3066375

TIBSHIRANI, R. J., RINALDO, A., TIBSHIRANI, R. and WASSERMAN, L. (2015). Uniform asymptotic inference and the bootstrap after model selection. Preprint. Available at arXiv:1506.06266.

VAN DE GEER, S., BÜHLMANN, P., RITOV, Y. and DEZEURE, R. (2013). On asymptotically optimal confidence regions and tests for high-dimensional models. Preprint. Available at arXiv:1303.0518.

WEINSTEIN, A., FITHIAN, W. and BENJAMINI, Y. (2013). Selection adjusted confidence intervals with more power to determine the sign. *J. Amer. Statist. Assoc.* **108** 165–176. MR3174610

ZHANG, C.-H. and ZHANG, S. S. (2014). Confidence intervals for low dimensional parameters in high dimensional linear models. *J. R. Stat. Soc. Ser. B. Stat. Methodol.* **76** 217–242. MR3153940

ZHONG, H. and PRENTICE, R. L. (2008). Bias-reduced estimators and confidence intervals for odds ratios in genome-wide association studies. *Biostatistics* **9** 621–634.

ZOLLNER, S. and PRITCHARD, J. K. (2007). Overcoming the winner's curse: Estimating penetrance parameters from case-control data. *Am. J. Hum. Genet.* **80** 605–615.

ZOU, H. and HASTIE, T. (2005). Regularization and variable selection via the elastic net. *J. R. Stat. Soc. Ser. B. Stat. Methodol.* **67** 301–320. MR2137327